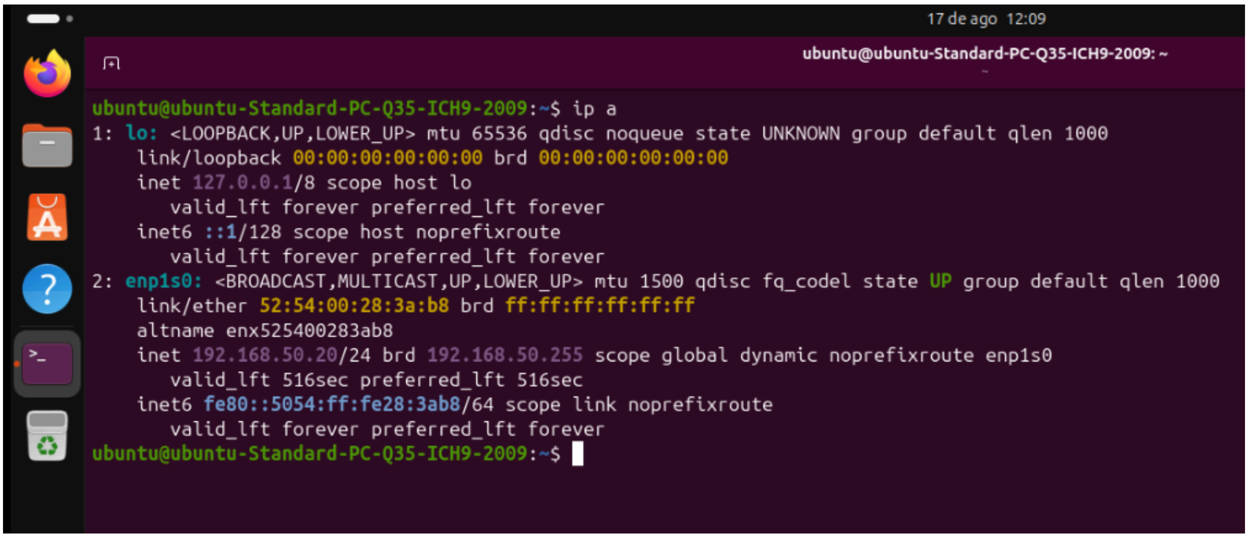
# Índice

# Cliente Ubuntu

## DHCP

As we can observe, the IP address **192.168.50.20** of the Ubuntu client is within the specified range **192.168.50.20 - 192.168.50.100**.




```
ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:28:3a:b8 brd ff:ff:ff:ff:ff:ff
    altname enx525400283ab8
    inet 192.168.50.20/24 brd 192.168.50.255 scope global dynamic noprefixroute enp1s0
       valid_lft 516sec preferred_lft 516sec
    inet6 fe80::5054:ff:fe28:3ab8/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009:~$
```

Now, we will make a reservation using the MAC address. We add the following lines to **/etc/dhcp/dhcpd.conf:**

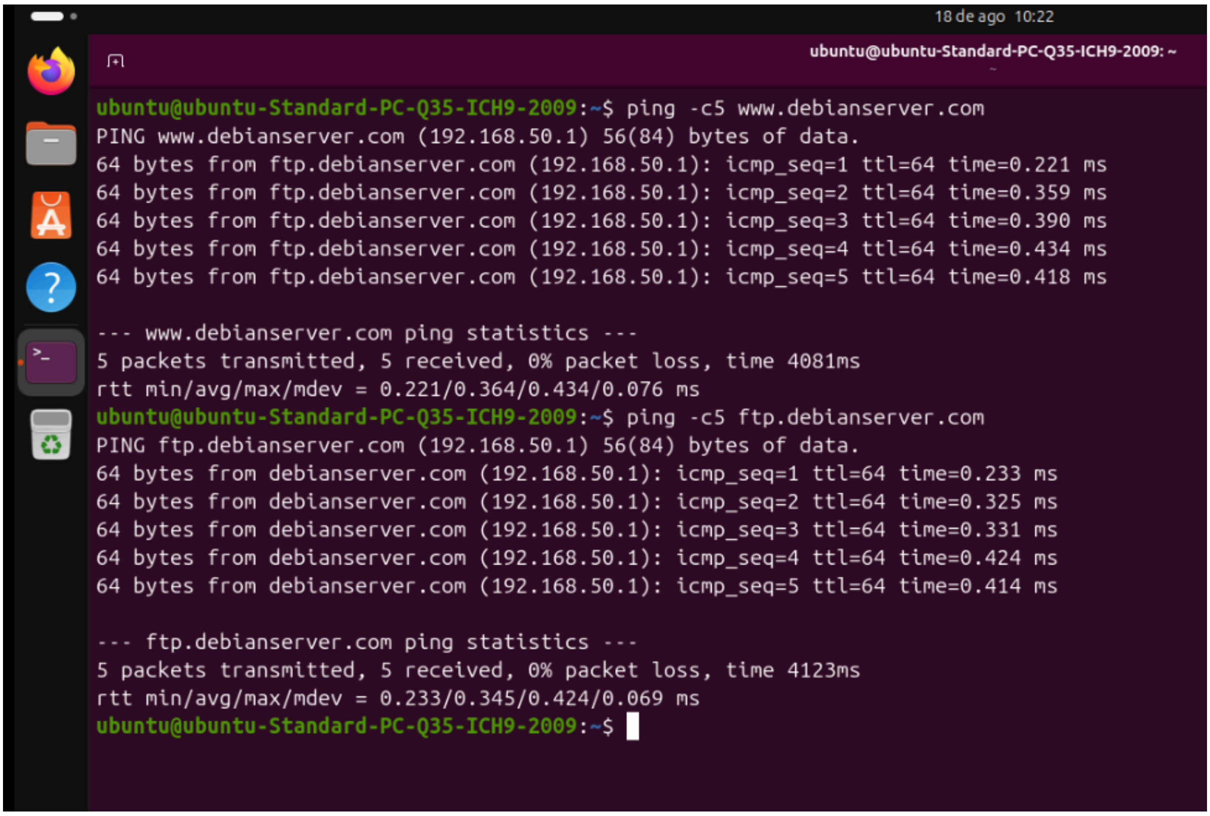
```
host ubuntu {
hardware ethernet 52:54:00:28:3a:b8;
fixed-address 192.168.50.31;
}
```

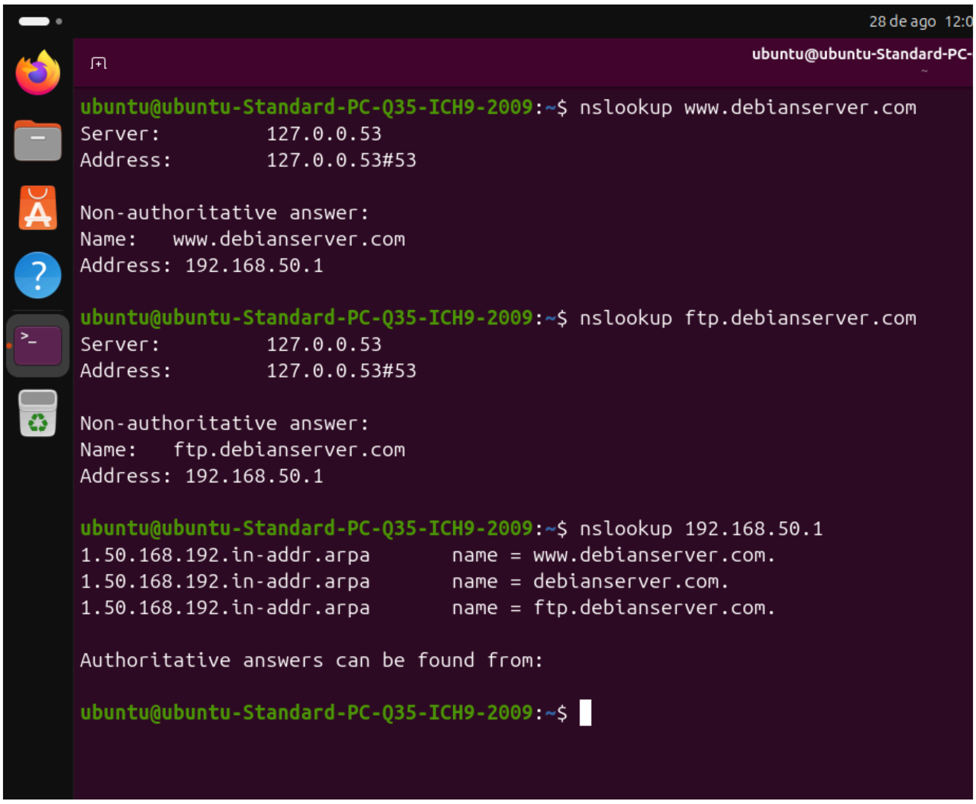
We restart Ubuntu and we can verify that the verification was successful.

```
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:28:3a:b8 brd ff:ff:ff:ff:ff:ff
    altname enx525400283ab8
    inet 192.168.50.31/24 brd 192.168.50.255 scope global dynamic noprefixroute enp1s0
       valid_lft 534sec preferred_lft 534sec
    inet6 fe80::5054:ff:fe28:3ab8/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
ubuntu@ubuntu-Standard-PC-Q35-ICH9-2009:~$
```

# DNS

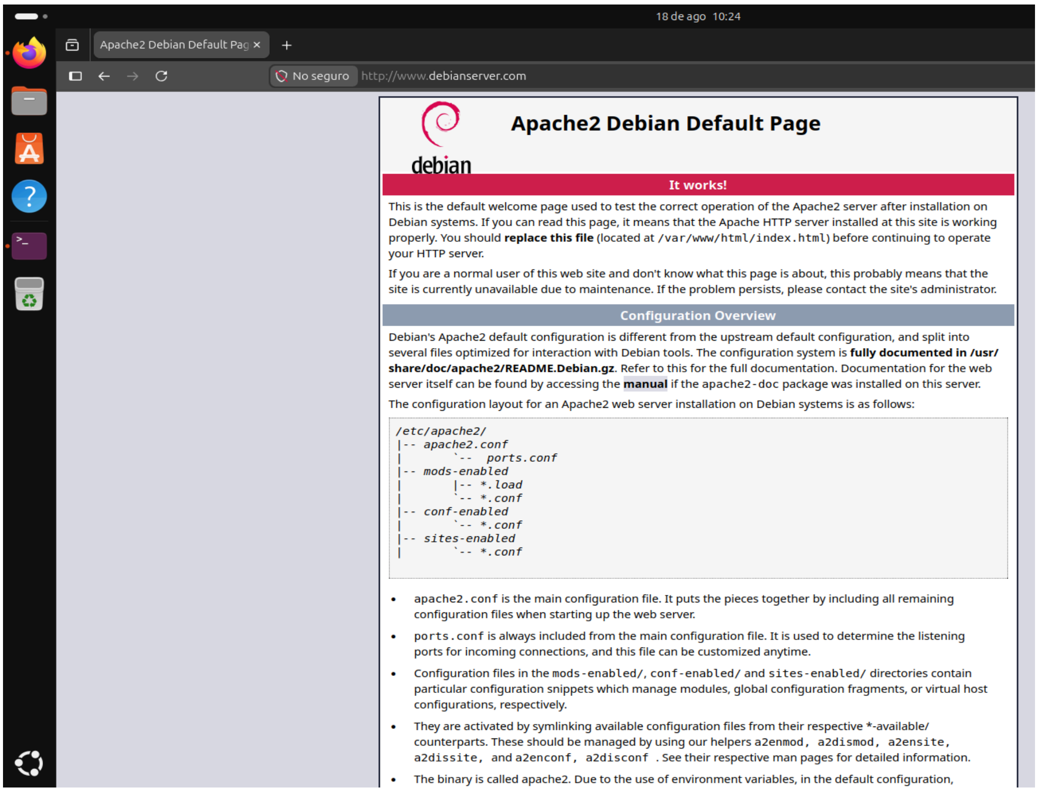
We ping the following domain names **www.debianserver.com, ftp.debianserver.com** and run **nslookup www.debianserver.com**, **nslookup ftp.debianserver.com**, and **nslookup 192.168.50.1.**



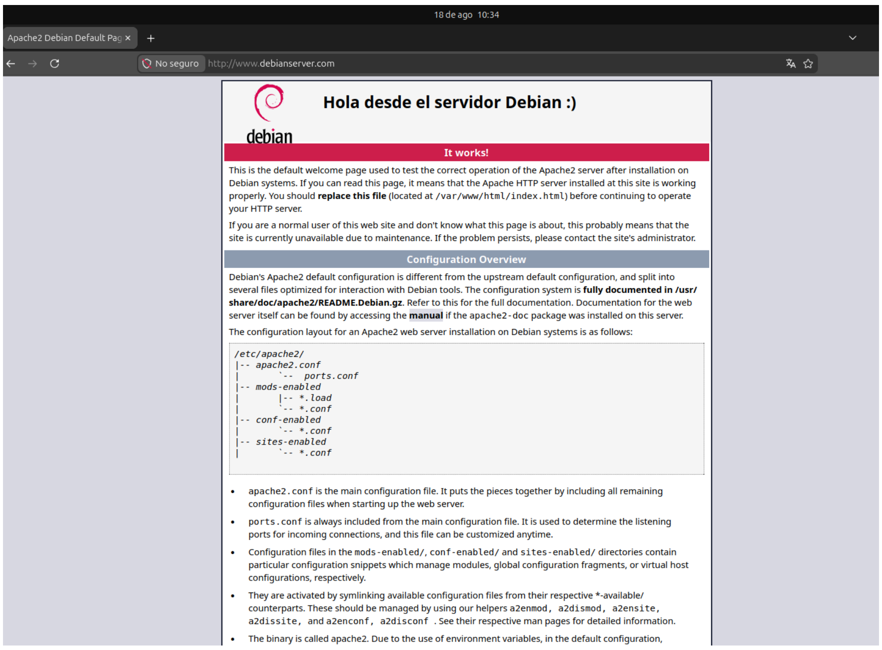
# Apache2

We open the browser and search for the following domain: **www.debianserver.com**




Now, we will modify the page header in **/var/www/html/index.html**

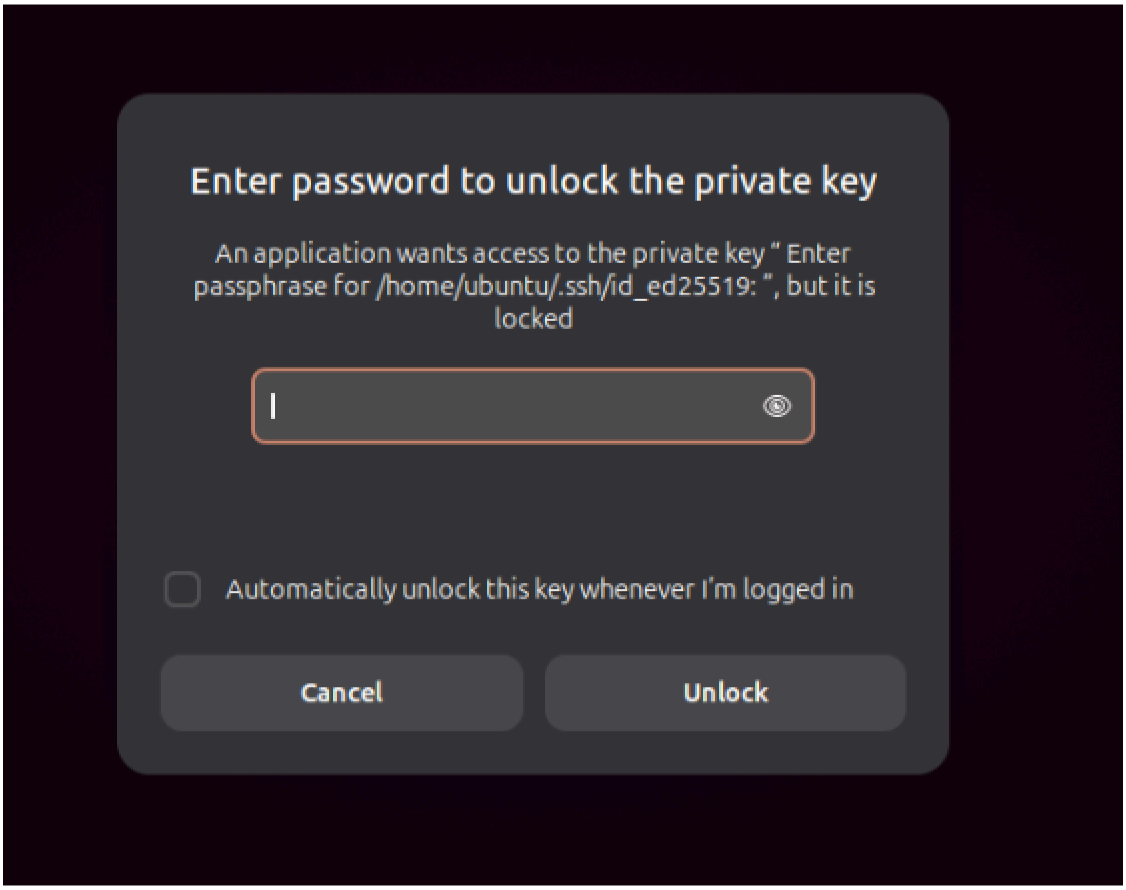

We reload the page on the Ubuntu client, and as we can observe, the changes took effect



# SSH

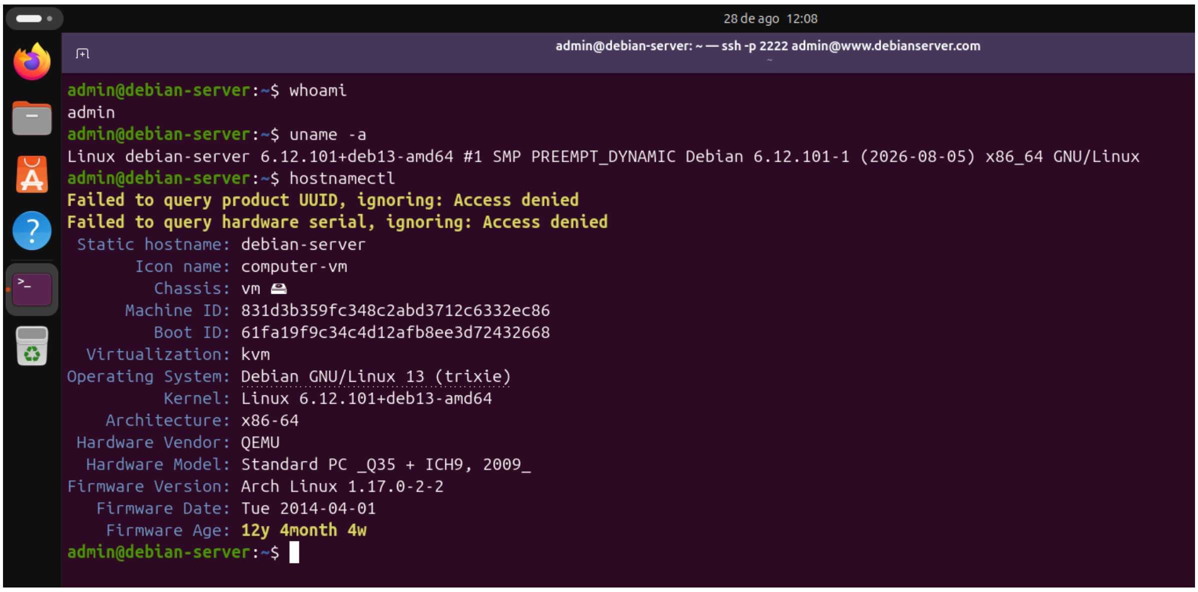
We connect "remotely" to the server via SSH. For this case, we have to create the SSH keys on the Ubuntu client.

Now, we run the command **ssh -p 2222 admin@www.debianserver.com** and it will prompt for the public key passphrase of the Ubuntu client.



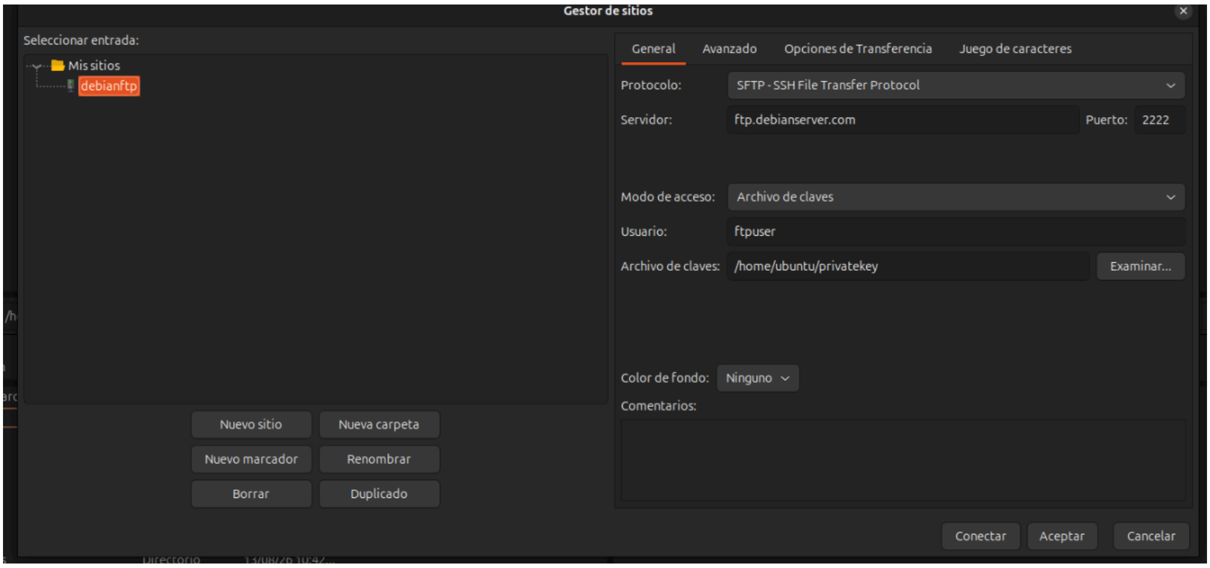

We provide the public key and it will grant us access to the server.

As we can observe, we indeed have "remote" access to the Debian server.

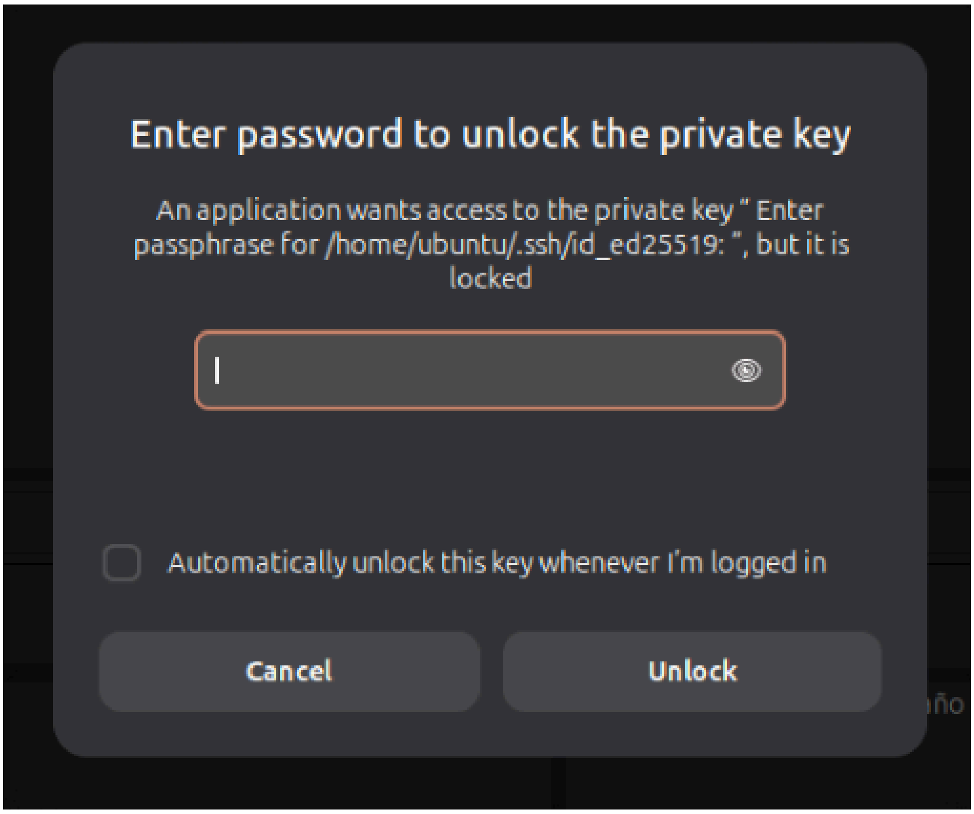


# SFTP

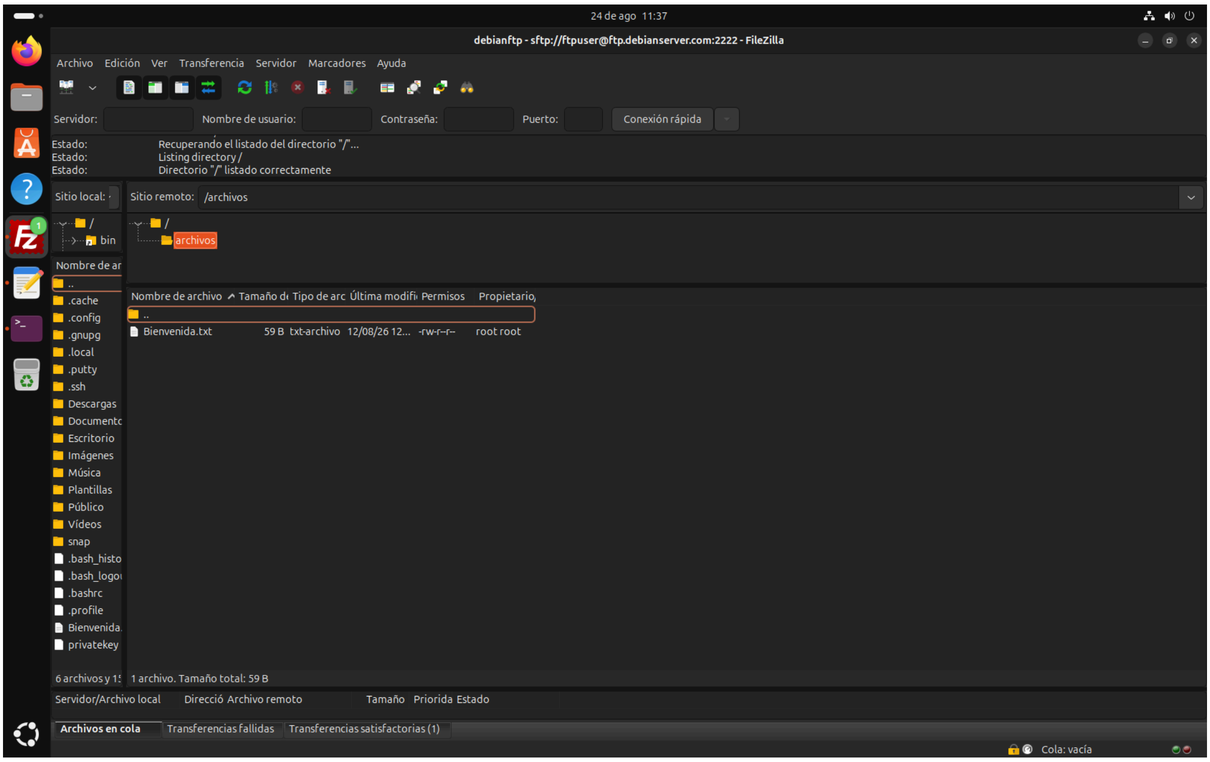
Now, using FileZilla, we create a new site with the following credentials and press the Connect button.



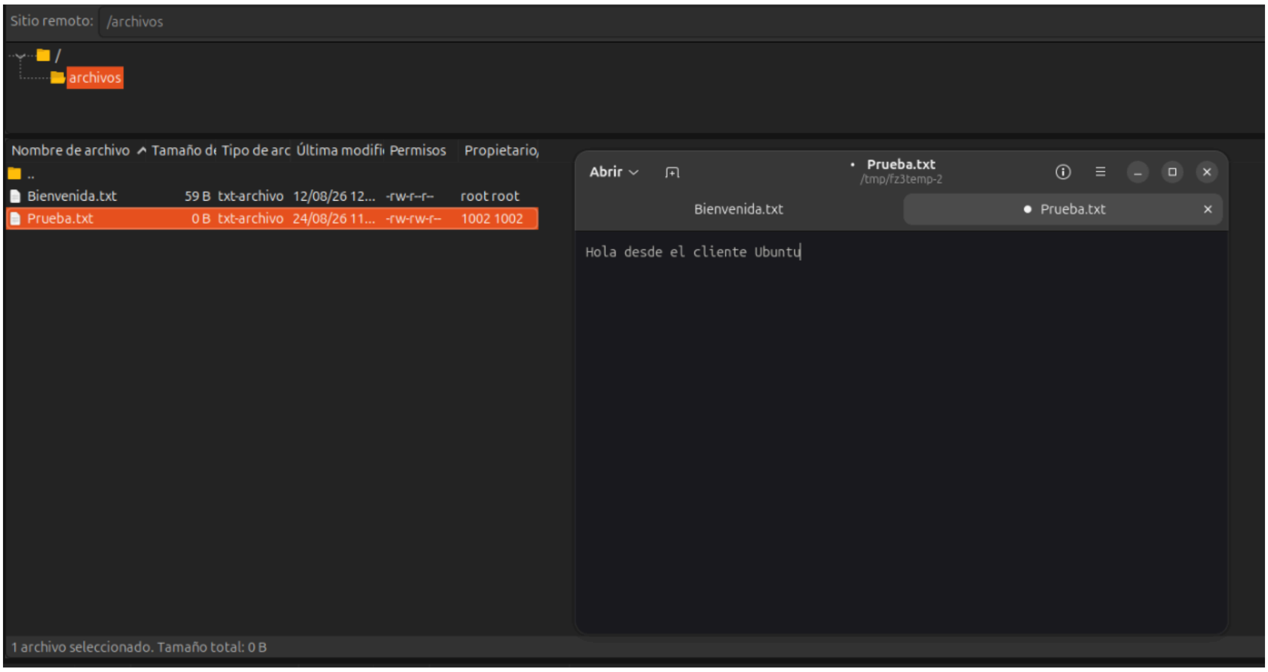

It will ask for our private key. Therefore, we select and load it.



We now have access to the 0+5 Array via SFTP. As we can observe, the first file we see is **Bienvenida.txt**




We proceed to add a file from the Ubuntu client, which will be named **Prueba.txt** and will contain “Hola desde el cliente Ubuntu”



As we can observe, the file and its content are reflected on the disk with RAID 50 array on the Debian server.

```
root@debian-server:/mnt/raid50/compartido/archivos# pwd
/mnt/raid50/compartido/archivos
root@debian-server:/mnt/raid50/compartido/archivos# cat Prueba.txt
Hola desde el cliente Ubuntu
root@debian-server:/mnt/raid50/compartido/archivos#
```

# Cliente Windows 10

## DHCP

As we can observe, the DHCP server assigned the IP **192.168.50.21** .which is within the range **192.168.50.20 - 192.168.50.100**

```
Adaptador de Ethernet Ethernet:

   Sufijo DNS específico para la conexión. . :
   Descripción . . . . . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Dirección física. . . . . . . . . . . . . : 52-54-00-6E-B0-D1
   DHCP habilitado . . . . . . . . . . . . . : sí
   Configuración automática habilitada . . . : sí
   Vínculo: dirección IPv6 local. . . : fe80::61bb:98c7:5024:4131%4(Preferido)
   Dirección IPv4. . . . . . . . . . . . . . : 192.168.50.21(Preferido)
   Máscara de subred . . . . . . . . . . . . : 255.255.255.0
   Concesión obtenida. . . . . . . . . . . . : martes, 25 de agosto de 2026 10:02:18
   La concesión expira . . . . . . . . . . . : martes, 25 de agosto de 2026 10:12:17
   Puerta de enlace predeterminada . . . . . : 192.168.50.1
   Servidor DHCP . . . . . . . . . . . . . . : 192.168.50.1
   IAID DHCPv6 . . . . . . . . . . . . . . . : 106058752
   DUID de cliente DHCPv6. . . . . . . . . . : 00-01-00-01-32-0F-47-41-52-54-00-6E-B0-D1
   Servidores DNS. . . . . . . . . . . . . . : 192.168.50.1
   NetBIOS sobre TCP/IP. . . . . . . . . . . : habilitado
PS C:\Users\Win10cliente>
```

Now we will make a reservation using the MAC address, placing the following text in **/etc/dhcp/dhcpd.conf** on the Debian server

```
host win {
hardware ethernet 52:54:00:6E:B0:D1;
fixed-address 192.168.50.41;
}
```

Now we execute the command **ipconfig /renew** on the Windows 10 Client, and we can observe that the reservation was executed successfully.

```
Windows PowerShell
PS C:\Users\Win10cliente> ipconfig /renew

Configuración IP de Windows


Adaptador de Ethernet Ethernet:

   Sufijo DNS específico para la conexión. . :
   Vínculo: dirección IPv6 local. . . : fe80::61bb:98c7:5024:4131%4
   Dirección IPv4. . . . . . . . . . . . . . : 192.168.50.41
   Máscara de subred . . . . . . . . . . . . : 255.255.255.0
   Puerta de enlace predeterminada . . . . . : 192.168.50.1
PS C:\Users\Win10cliente>
```

# DNS

We execute the commands **www.debianserver.com y ftp.debianserver.com**. Afterwards, we execute the commands **nslookup www.debianserver.com** , **nslookup ftp.debianserver.com** and **nslookup 192.168.50.1**

Windows PowerShell

```
PS C:\Users\Win10cliente> ping www.debianserver.com

Haciendo ping a www.debianserver.com [192.168.50.1] con 32 bytes de datos:
Respuesta desde 192.168.50.1: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.50.1: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.50.1: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.50.1: bytes=32 tiempo<1m TTL=64

Estadísticas de ping para 192.168.50.1:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 0ms, Máximo = 0ms, Media = 0ms
PS C:\Users\Win10cliente> ping ftp.debianserver.com

Haciendo ping a ftp.debianserver.com [192.168.50.1] con 32 bytes de datos:
Respuesta desde 192.168.50.1: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.50.1: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.50.1: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.50.1: bytes=32 tiempo<1m TTL=64

Estadísticas de ping para 192.168.50.1:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 0ms, Máximo = 0ms, Media = 0ms
PS C:\Users\Win10cliente> _
```

Windows PowerShell

```
Windows PowerShell
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Prueba la nueva tecnología PowerShell multiplataforma https://aka.ms/pscore6

PS C:\Users\Win10cliente> nslookup www.debianserver.com
Servidor:  www.debianserver.com
Address:  192.168.50.1

Nombre:  www.debianserver.com
Address:  192.168.50.1

PS C:\Users\Win10cliente> nslookup ftp.debianserver.com
Servidor:  ftp.debianserver.com
Address:  192.168.50.1

Nombre:  ftp.debianserver.com
Address:  192.168.50.1

PS C:\Users\Win10cliente> nslookup 192.168.50.1
Servidor:  debianserver.com
Address:  192.168.50.1

Nombre:  ftp.debianserver.com
Address:  192.168.50.1

PS C:\Users\Win10cliente>
```
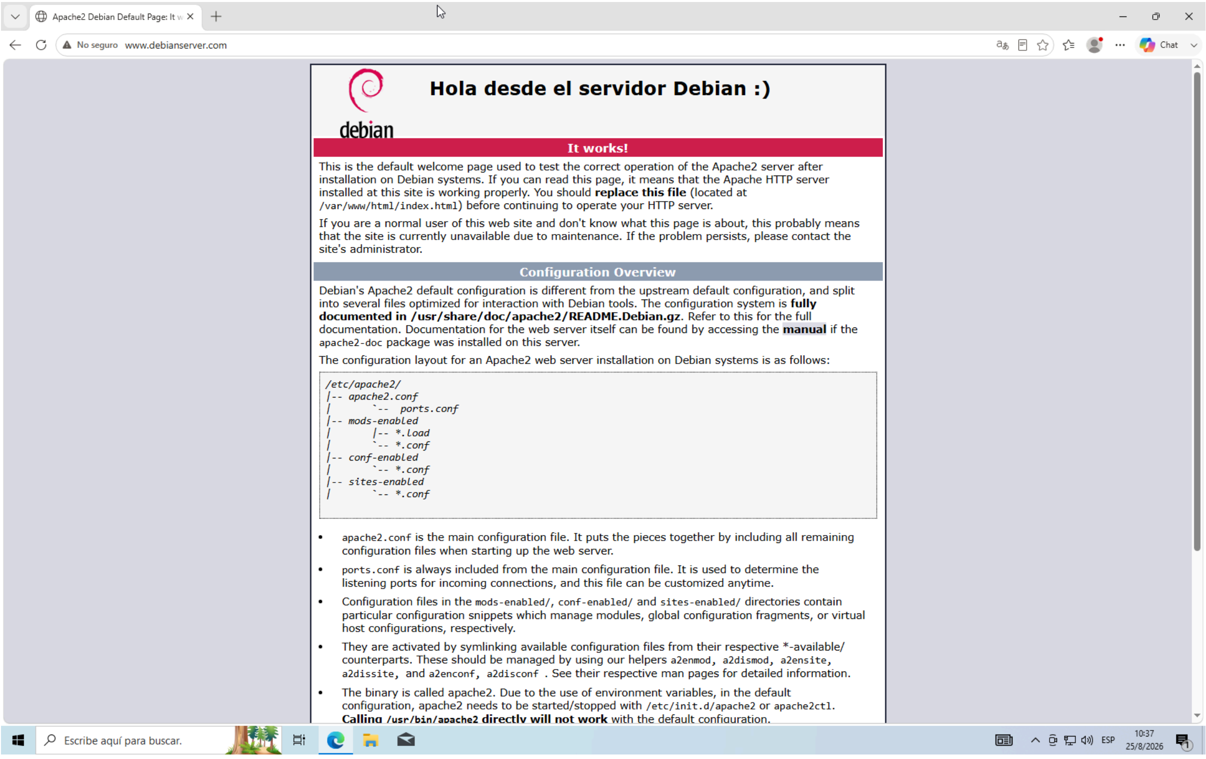

# Apache2

We enter the browser and type the address **www.debianserver.com**, As we can observe, the Apache2 service works perfectly.



# SSH

We access the server using a cryptographic key.

```
PS C:\Users\Win10cliente> ssh -p 2222 admin@www.debianserver.com
Authorized access only. All activity is monitored and logged
Enter passphrase for key 'C:\Users\Win10cliente/.ssh/id_ed25519':
Linux debian-server 6.12.101+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.101-1 (2026-08-05) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Aug 25 13:57:57 2026 from 192.168.50.41
```

admin@debian-server: ~

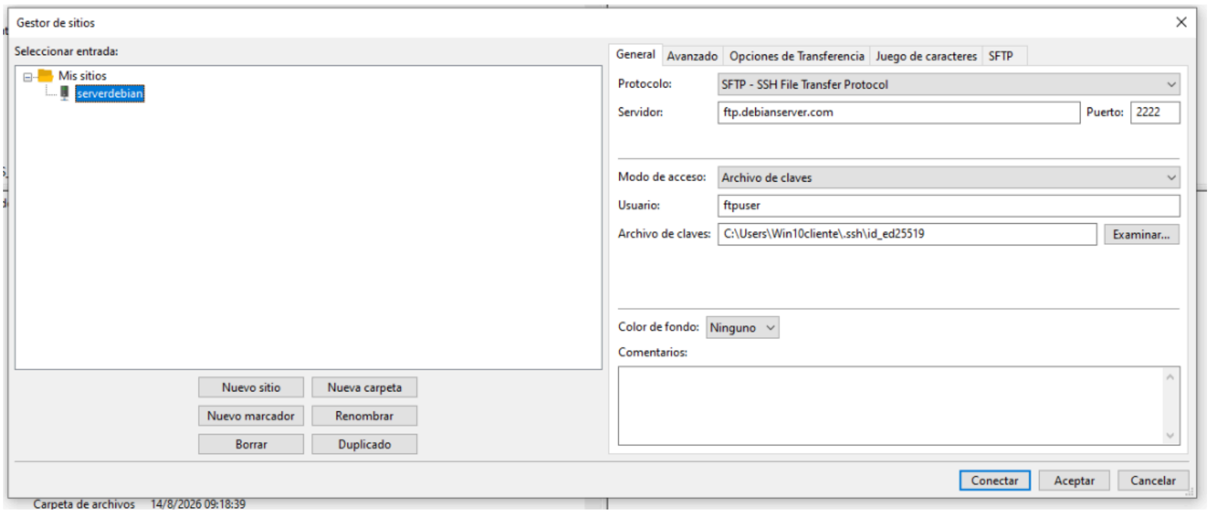
```
admin@debian-server:~$ hostnamectl
Failed to query product UUID, ignoring: Access denied
Failed to query hardware serial, ignoring: Access denied
 Static hostname: debian-server
       Icon name: computer-vm
         Chassis: vm ⍰
      Machine ID: 831d3b359fc348c2abd3712c6332ec86
         Boot ID: 61fa19f9c34c4d12afb8ee3d72432668
  Virtualization: kvm
Operating System: Debian GNU/Linux 13 (trixie)
          Kernel: Linux 6.12.101+deb13-amd64
    Architecture: x86-64
 Hardware Vendor: QEMU
  Hardware Model: Standard PC _Q35 + ICH9, 2009_
Firmware Version: Arch Linux 1.17.0-2-2
   Firmware Date: Tue 2014-04-01
    Firmware Age: 12y 4month 3w 6d
admin@debian-server:~$
```

admin@debian-server: ~

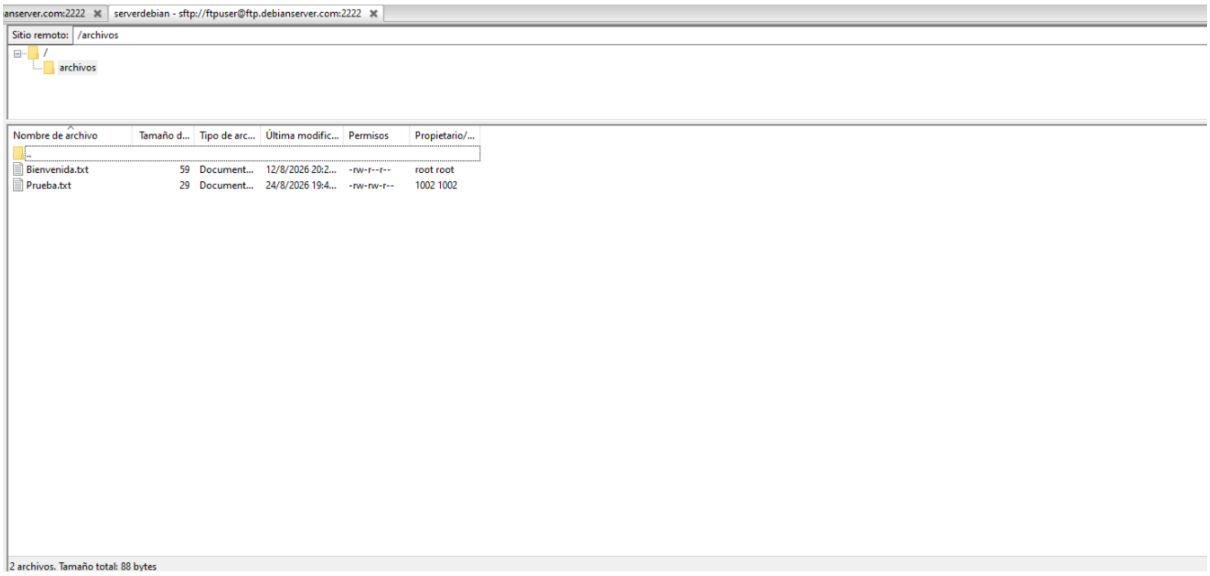
```
admin@debian-server: $ whoami
admin
admin@debian-server: $ uname -a
Linux debian-server 6.12.101+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.101-1 (2026-08-05) x86_64 GNU/Linux
admin@debian-server: $
```

# SFTP

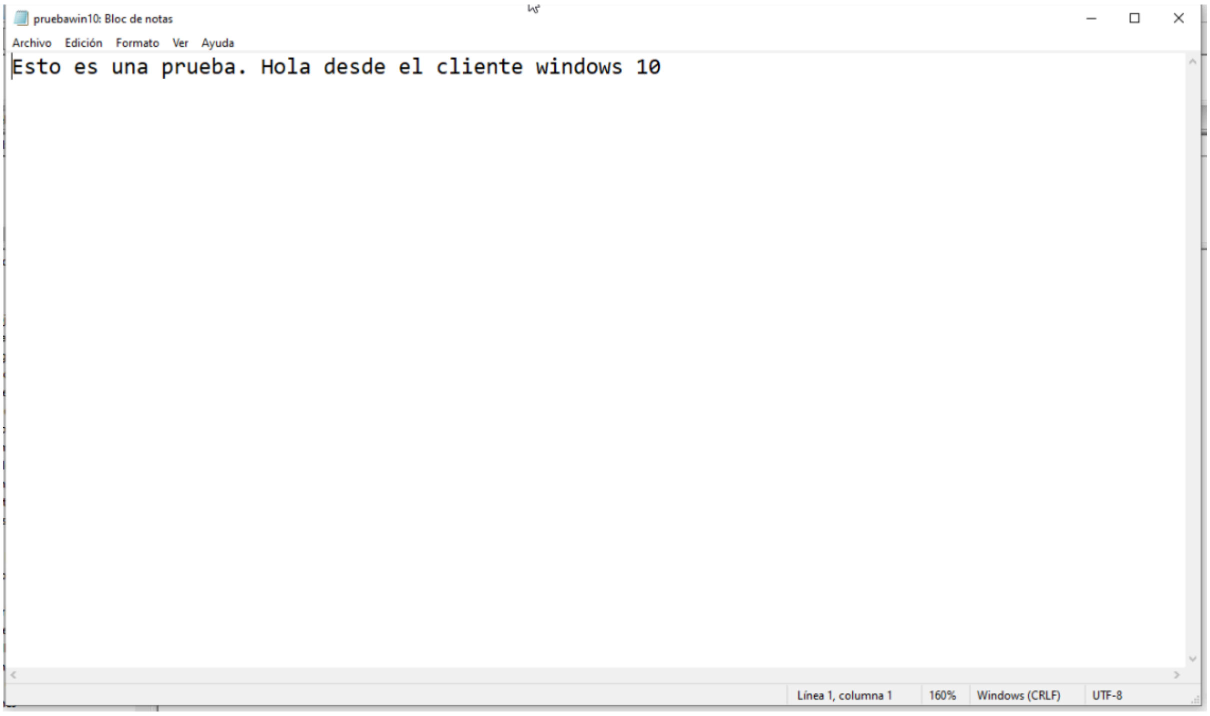
We export our private key to FileZilla for SFTP access. We enter the required parameters into the corresponding fields:




As we can observe, we already have access via SFTP.



Now, we will upload a test file named **pruebawin10.txt** with the following content:




The file **pruebawin10.txt** was successfully uploaded to the Debian Server , specifically to the RAID 50 array.